Airlines Management System - Implementation Report

Project: Salesforce Capstone Project - Airlines Management System
Institution: LNCT Group of Colleges
Program: Salesforce COE Skilling Program (Wipro)
Date: October 21, 2026
Org: chandanpathak2005.fae1d6e8e465@agentforce.com

#

Executive Summary

This document provides a comprehensive overview of the Airlines Management System implementation in Salesforce. The project manages flight operations, schedules, pilots, and operational workflows with automated notifications and role-based access controls.

#

1. Application Setup

##

Lightning Application

- **Application Name:** Airlines Management System
- **Status:** Configured and deployed
- **Navigation Items:** Flight, Flight Schedule, Pilot, Pilot Schedule

##

Custom Objects Created

1. **Flight__c** - Flight master data
2. **Flight_Schedule__c** - Flight route and timing details
3. **Pilot__c** - Pilot profile and experience
4. **Pilot_Schedule__c** - Pilot assignments to flights

#

---

2. Data Model Implementation

##

2.1 Flight Object (Flight__c)

Field Name	Data Type	Description
Name (Flight ID)	Auto Number	Format: FL-{0000}
Flight_Name__c	Text	Example: GO-102

| Flight_Company__c | Picklist | IndiGo / Air India / GoAir / Kingfisher |

Relationships:
- One-to-Many relationship with Flight_Schedule__c (Master-Detail)

##

2.2 Flight Schedule Object (Flight_Schedule__c)

Field Name	Data Type	Description
Flight__c	Master-Detail	Relationship to Flight
Source_Name__c	Picklist	Source city
Destination_Name__c	Picklist	Destination city
Departure_Date_Time__c	DateTime	Scheduled departure
Arrival_Date_Time__c	DateTime	Scheduled arrival
Duration__c	Formula (Text)	Calculate journey duration
Status__c	Picklist	Open / In Progress / Closed / Cancelled
First_Pilot__c	Lookup	First pilot assignment
Second_Pilot__c	Lookup	Second pilot assignment

Formula Fields:
```apex
```

Duration__c = TEXT(FLOOR((Arrival_Date_Time__c - Departure_Date_Time__c) * 24)) & " hours " & TEXT(ROUND(MOD((Arrival_Date_Time__c - Departure_Date_Time__c) * 24 * 60, 60), 0)) & " minutes"
```

**Business Rules:**
- Default status: "Open" when created
- Status transitions: Open â†' In Progress â†' Closed
- Source and destination must be different
- Departure must be before arrival

---

##

---

### 2.3 Pilot Object (Pilot__c)

| Field Name | Data Type | Description |
|---|---|---|
| First_Name__c | Text | Mandatory |
| Last_Name__c | Text | Pilot surname |
| Date_of_Birth__c | Date | Date of birth |
| Age__c | Formula (Number) | Calculated from DOB |
| Contact_Number__c | Phone | Pilot mobile number |
| Email_Address__c | Email | Mandatory |
| Experience__c | Picklist | Less than 1 year / 2-5 years / More than 5 years |
| Level__c | Formula (Text) | Junior / Senior / Super Senior |
```

Formula Fields:

Age Calculation:

```apex
Age__c = FLOOR((TODAY() - Date_of_Birth__c) / 365.25)
```

Level Calculation:

```apex
Level__c = IF(TEXT(Experience__c) = "Less than 1 year",
"Junior",
IF(TEXT(Experience__c) = "2-5 years", "Senior",
IF(TEXT(Experience__c) = "More than 5 years", "Super
Senior", "")))
```

Business Rules:

- First name and last name must be different
- Minimum age: 18 years
- Email and contact number mandatory

##

2.4 Pilot Schedule Object (Pilot_Schedule__c)

| Field Name | Data Type | Description |

|------------|-----------|-------------|
Flight_Schedule__c	Master-Detail	Relationship to Flight Schedule
Pilot_1__c	Lookup	First pilot
Pilot_2__c	Lookup	Second pilot

Business Rules:
- Pilot 1 and Pilot 2 must be different
- Email notification sent to pilots on assignment

#

---

3. Validation Rules Implemented

##

3.1 Flight_Schedule__c.Departure_Before_Arrival

```xml

Departure_Date_Time__c >= Arrival_Date_Time__c


Departure date/time must be earlier than arrival date/time.

```

##

3.2 Flight_Schedule__c.Different_Cities

```xml

Source_Name__c = Destination_Name__c

Source and destination cities must be different.

```

##

3.3 Flight_Schedule__c.Distinct_Pilots (on related schedules)

```xml

First_Pilot__c = Second_Pilot__c

First Pilot and Second Pilot must be different.

```

##

3.4 Pilot_Schedule__c.Distinct_Pilots

```xml
```

Pilot_1__c = Pilot_2__c

Pilot 1 and Pilot 2 must be different pilots.

```

##

---

### 3.5 Pilot__c.Minimum_Age

```xml

**Age__c < 18**

**Pilot must be at least 18 years old.**

```

##

3.6 Pilot__c.Different_Names

```xml

**LOWER(TEXT(First_Name__c)) =**
**LOWER(TEXT(Last_Name__c))**

First name and last name must not be identical.

```

#

---

4. Apex Development

##

4.1 FlightScheduleTriggerHandler Class

Purpose: Prevent duplicate active schedules and maintain data integrity

File Location: `force-app/main/default/classes/FlightScheduleTriggerHandler.cls`

Key Methods:

1. **validateAndUpdate()** - Main validation and status update logic
- Prevents duplicate schedules for same flight and route/time
- Auto-updates status based on departure/arrival times
- Validates data integrity
```

2. **setStatusIfAvailable() - Helper method for safe status updates**

Business Logic:
```apex
// Auto-status updates:
// - If arrival time passed â†' Status = "Closed"
// - If departure time passed â†' Status = "In Progress"
// - If new record â†' Status = "Open"
```

Duplicate Prevention:
- Checks for existing schedules with same Flight, Source, Destination, and Departure Time
- Prevents creation/update if duplicate found
- Error message: "An active schedule already exists for this flight, route, and time."

##

4.2 AirlinesSOQLService Class

Purpose: Demonstrate required SOQL queries including relationship queries

File Location: `force-app/main/default/classes/AirlinesSOQLService.cls`

###

Parent-to-Child Relationship Query

Method: `getFlightWithSchedules(Id flightId)`

```apex
SELECT Id, Name, Flight_Name__c, Flight_Company__c,
(SELECT Id, Name, Source_Name__c, Destination_Name__c,
Departure_Date_Time__c, Arrival_Date_Time__c,
Duration__c, Status__c
FROM Flight_Schedules__r
ORDER BY Departure_Date_Time__c DESC)
FROM Flight__c
WHERE Id = :flightId
```

Purpose: Retrieve a Flight record with all its related Flight Schedule records in a single query.

###

Child-to-Parent Relationship Query

Method: `getSchedulesWithFlightInfo()`

```apex
SELECT Id, Name, Source_Name__c, Destination_Name__c,
Departure_Date_Time__c, Arrival_Date_Time__c, Status__c,
Flight__c, Flight__r.Name, Flight__r.Flight_Name__c,
Flight__r.Flight_Company__c
FROM Flight_Schedule__c
ORDER BY Departure_Date_Time__c DESC
LIMIT 100
```

Purpose: Retrieve Flight Schedule records with parent Flight information.

###

Additional SOQL Queries

1. **getSchedulesByFlight(Id flightId)**
- Retrieves all schedules for a specific flight

2. **getSchedulesByStatus(String status)**
- Retrieves schedules by status (Open, Cancelled, etc.)

3. **getSchedulesBySourceCity(String sourceCity)**
- Retrieves schedules departing from a specific city

4. **getSchedulesByDateRange(DateTime startDate,

DateTime endDate)**

- Retrieves schedules within a date range

#

---

5. Testing Evidence

##

5.1 Validation Rule Tests

****Test Case 1: Departure Before Arrival****

- **Input: Departure: 2026-10-25 10:00 AM, Arrival: 2026-10-25 09:00 AM**

- **Expected: Error message displayed**

- **Result: âœ… Validation triggered, record creation blocked**

****Test Case 2: Distinct Pilots****

- **Input: First Pilot = Pilot A, Second Pilot = Pilot A**

- **Expected: Error message displayed**

- **Result: âœ… Validation triggered, record creation blocked**

****Test Case 3: Minimum Age****

- **Input: Date of Birth: 2015-01-01 (Age: 11 years)**

- **Expected: Error message displayed**

- **Result:** âœ… Validation triggered, record creation blocked

Test Case 4: Different Names
- **Input:** First Name: "John", Last Name: "John"
- **Expected:** Error message displayed
- **Result:** âœ… Validation triggered, record creation blocked

---

##

5.2 Apex Trigger Tests

Test Case 1: Duplicate Schedule Prevention
- **Setup:** Create a schedule for Flight GO-102, Delhi â†' Mumbai, Departure: 2026-10-25 10:00 AM
- **Test:** Attempt to create another schedule with same details
- **Expected:** Error: "An active schedule already exists..."
- **Result:** âœ… Duplicate blocked successfully

Test Case 2: Status Auto-Update
- **Setup:** Create schedule with departure time in the past
- **Test:** Insert record
- **Expected:** Status automatically set to "In Progress" or "Closed"
- **Result:** âœ… Status updated automatically

##

5.3 SOQL Query Tests

Test Case 1: Parent-to-Child Query

```apex
List flights =
AirlinesSOQLService.getFlightWithSchedules(flightId);
System.assertEquals(1, flights.size());
System.assert(flights[0].Flight_Schedules__r.size() > 0);
```

Result: âœ… Query returns Flight with related Schedules

Test Case 2: Child-to-Parent Query

```apex
List schedules =
AirlinesSOQLService.getSchedulesWithFlightInfo();
System.assert(schedules[0].Flight__r != null);
System.assertNotEquals(null,
schedules[0].Flight__r.Flight_Name__c);
```

Result: âœ… Query returns Schedules with parent Flight information

---

#

6. Deployment Status

##

Successfully Deployed Components

Component Type	Component Name	Status
Custom Object	Flight__c	âœ… Deployed

| Custom Object | Flight_Schedule__c | âœ… Deployed |

| Custom Object | Pilot__c | âœ… Deployed |

| Custom Object | Pilot_Schedule__c | âœ… Deployed |

| Validation Rule | Departure_Before_Arrival | âœ… Deployed |

| Validation Rule | Distinct_Pilots (x2) | âœ… Deployed |

| Validation Rule | Minimum_Age | âœ… Deployed |

| Validation Rule | Different_Names | âœ… Deployed |

| Formula Field | Duration__c | âœ… Deployed |

| Formula Field | Age__c | âœ… Deployed |

| Formula Field | Level__c | âœ… Deployed |

| Apex Class | FlightScheduleTriggerHandler

| âœ… Deployed |

| Apex Class | AirlinesSOQLService | âœ… Deployed |

#

---

7. Remaining Requirements

##

High Priority

1. **Record-Triggered Flows**
- Flight scheduling notifications
- Status change notifications (departure, arrival)
- Cancellation alerts to admin and pilots
- Pilot assignment email notifications

2. **Users and Security**
- Create 6 users: CEO, Anna Hajare, Rajesh Deshpande,

Neha Kakkad, Anu Malik, Katappa, Bahubali

- Configure role hierarchy
- Setup profiles and permission sets
- Configure OWD (Organization-Wide Defaults)
- Create sharing rules for Mumbai and Pune groups

3. **Reports and Dashboards**

- Dashboard: Flights scheduled for today

- Dashboard: Flight schedules by month
- Pie chart: Closed vs Cancelled flights
- Report: All schedules for each flight
- Report: Flight schedules in current month
- Report: Flight schedules over last 3 months

#

8. Technical Architecture

##

Object Relationships

```

# Flight__c (Master)

# â””â”€â”€ Flight_Schedule__c

# (Detail)

# â”œâ”€â”€ First_Pilot__c â†’

# Pilot__c (Lookup)

# â””â”€â”€ Second_Pilot__c â†’

# Pilot__c (Lookup)
```

Pilot_Schedule__c (Junction Object)
â”œâ”€â”€
Flight_Schedule__c (Master-Detail)
â”œâ”€â”€ Pilot_1__c â†’ Pilot__c (Lookup)
â””â”€â”€ Pilot_2__c â†’ Pilot__c (Lookup)
```

##

---

# Data Flow
```

1. **Flight Creation** â†’ Flight record created
2. **Schedule Planning** â†’ Flight Schedule created (validates source â‰ destination, departure < arrival)
3. **Pilot Assignment** â†’ Pilots assigned to schedule (validates distinct pilots, age â‰¥ 18)
4. **Status Updates** â†’ Automatic status transitions based on date/time

5. **Notifications** â†’ (To be implemented via Flows)

#

---

9. Code Quality and Best Practices

##

Implemented Best Practices âœ… Apex classes follow naming conventions

- âœ… Error handling with addError() method
- âœ… SOQL queries use bind variables to prevent injection
- âœ… Bulkified trigger handler (processes lists, not single records)
- âœ… Validation rules use formula expressions
- âœ… Formula fields calculate derived values
- âœ… Relationship queries demonstrate parent-child navigation

##

Security Considerations

âœ… Validation rules prevent invalid data

âœ… Trigger prevents duplicate records

âœ… Field-level validation for age, names, dates

âœ… Relationship integrity maintained

#

---

10. Summary and Conclusion

##

Completed Tasks âœ…

- âœ… Custom objects with all required fields
- âœ… Relationships (Master-Detail and Lookup)
- âœ… 6 Validation rules for data integrity
- âœ… 3 Formula fields for

calculations

- âœ… Apex Trigger Handler for duplicate prevention
- âœ… SOQL Service with relationship queries
- âœ… Parent-to-Child and Child-to-Parent queries
- âœ… Test scenarios documented

##

Pending Tasks ðŸ”„

- ðŸ”„ Record-Triggered

Flows for email notifications

- ðŸ”„ User setup and role hierarchy

- ðŸ”„ Security configuration (OWD, Sharing Rules)

- ðŸ”„ Reports and Dashboards

- ðŸ”„ Sample data creation for demonstrations

##

Compliance with Requirements

This implementation addresses **Project 01 - Airlines Management System** requirements from the LNCT Salesforce Capstone Project Brief, specifically:

- âœ… Data model and relationships
- âœ… Business rules via validation
- âœ… Formula fields for calculations
- âœ… Apex Trigger requirement

- âœ… SOQL queries (parent-child, child-parent, filters)
- ðŸ”„ Automation (Flows - pending)
- ðŸ”„ Security (Users, Roles - pending)
- ðŸ”„ Reports/Dashboards (pending)

#

---

11. File Locations

##

Apex Classes

- `force-app/main/default/classes/FlightSch

- `force-app/main/default/classes/FlightSch

- `force-app/main/default/classes/AirlinesS(

##

Objects

- `force-

app/main/default/objects/Flight__c/

- `force-app/main/default/objects/Flight_Sc

- `force-app/main/default/objects/Pilot__c/`

- `force-app/main/default/objects/Pilot_Sch

##

Validation Rules

- `force-app/main/default/objects/Flight_Sc

meta.xml`

- `force-app/main/default/objects/Flight_Sc
meta.xml`
- `force-app/main/default/objects/Pilot_Sch
meta.xml`
- `force-app/main/default/objects/Pilot__c/
meta.xml`
- `force-app/main/default/objects/Pilot__c/
meta.xml`

#

---

12. Contact and Support

Project Team: LNCT Group of Colleges

Program: Salesforce COE Skilling Program

Partner: Wipro

Org URL: https://orgfarm-a440bc4828-dev-ed.develop.my.salesforce.com

---

Document Version: 1.0

Last Updated: October 21, 2026

Status: Development Phase - Core Implementation Complete

---

*This document serves as evidence of implementation

for the Salesforce Capstone Project evaluation.*